Leading University

Department of Computer Science and Engineering

CSE 3232

Lab Report 1

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Title:

Creating and configuring a network with 10 computer, 2 hub and testing the connectivity between them by sending message from one computer to another.

Objective:

- Implement peer-to-peer network using packet tracer
- How a hub can be used to create a network.
- Connecting two networks with the use of multiple hub

Background Theory:

A hub is a basic networking device used to connect multiple computers or devices in a network. It's important to note that hubs operate at the physical layer of the OSI model, which means they simply transmit signals to every device connected to them without any intelligence to manage where the data goes. When a device sends data through a hub, the hub broadcasts that data to all devices on the network. Only the device with the matching address processes the data, while all the others discard it.

However, hubs have their limitations. For one, they typically offer only a small number of ports, usually between 4 to 12. This can be an issue when trying to connect multiple devices, as the number of available ports can quickly run out. Hubs are also inefficient for large networks, as the broadcast nature of the device creates unnecessary traffic and can cause network slowdowns, especially when too many devices are connected. For bigger networks, switches are generally preferred over hubs because switches only send data to the intended recipient, saving bandwidth and improving performance.

Observation and Findings:

During the simulation, I connected 10 PCs using two hubs in Packet Tracer. Each hub was connected to 5 PCs, which made sense given that each hub had only 6 ports. The simulation picture is given below:

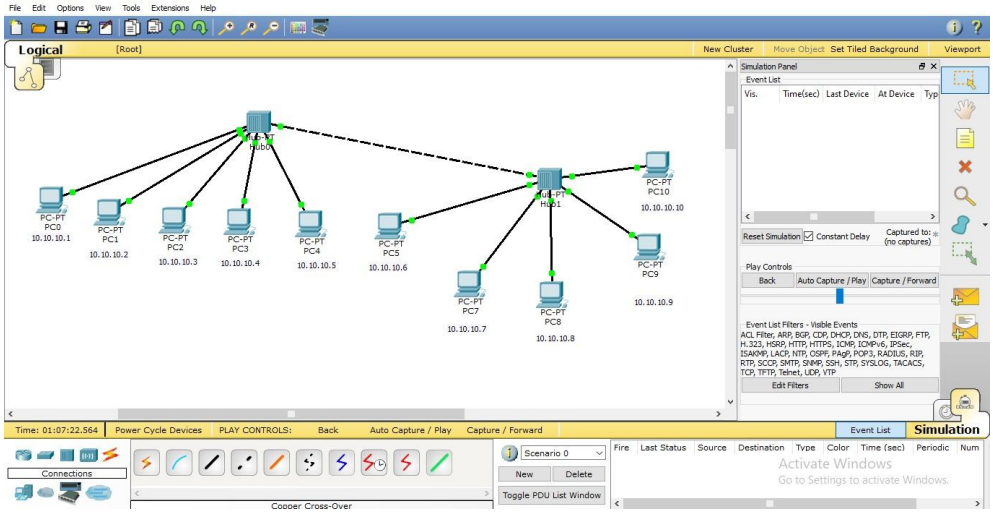

Figure 01: Network of 10 computer using hub

Some key observations were:

- **Hub Port Limits**: Since each hub only had 6 ports, it wasn't enough to connect all 10 PCs, so we had to use two hubs. This shows how hubs can be restrictive when we need to connect more devices.
- **Broadcasting Data**: Whenever one PC sent data, the hub broadcasted it to all the connected devices. But, as expected, only the destination PC processed the message, and the others just ignored it. This confirmed that hubs don't have the ability to target specific devices. They just send data out to everyone.
- **Network Behavior**: Even though all PCs received the data, only the PC that the message was intended for responded, which is typical for a hub-based network. The rest just discard it, creating unnecessary traffic.

Discussion:

This experiment really highlights some of the limitations of using hubs in a network. While hubs are simple devices that can easily connect a few computers, they're not the best choice for larger networks. Here's why:

- **Port Limitations**: Each hub had only 6 ports, which meant we had to use two hubs just to connect 10 PCs. If we had more devices, we'd need even more hubs, making the network unnecessarily complex. This just isn't practical for larger networks.
- **Inefficient Bandwidth Usage**: Because hubs send data to all connected devices, the network becomes flooded with unnecessary traffic. Only the correct PC processes the data, while all the others discard it. This wastes bandwidth and can slow things down, especially as the number of devices grows.
- **Collisions**: Hubs also operate in a shared network environment. This means that when multiple devices are transmitting at the same time, there's a high chance of collisions, which can slow down the network further as devices have to resend their data.

Conclusion:

The experiment confirmed that hubs are fine for small networks but aren't ideal for larger ones. In this case, we needed two hubs just to connect 10 PCs because each hub only had 6 ports. The fact that hubs broadcast data to all connected devices also leads to inefficient use of bandwidth and can create network congestion, especially as the network size increases.

To conclude, while hubs are simple to set up and work well in small networks, they come with some serious drawbacks when used in larger environments. For bigger, more complex networks, switches or routers should be used instead, as they're more efficient and designed to handle larger amounts of traffic without causing slowdowns.